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An Improved System for Detection of Tea Leaf Disease Using Hybrid AI Models

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Abstract

The Agricultural domain is segmented into various categories, such as grains, fruits, vegetables, and flowers. Leaf diseases are commonly found in all of these categories. Deep learning (DL) and Machine learning (ML) models are used to classify diseases and investigate the most recent advancements to identify plant leaf diseases. This paper presents three hybrid deep learning models with the anticipation of improved results. Three separate pre-trained deep learning models have been utilized and a hybrid approach is proposed. The pre-trained modes like CNN, VGG19, and ResNet50 are merged using transfer learning and global average pooling features to create three hybrid models which are the combination of VGG19 with global average pooling, VGG19, ResNet50, and CNN with the use of transfer learning and the third model is the merging of VGG19 with features of CNN model. The proposed models use transfer learning and global average pooling concepts to yield improved results. The three hybrid models are Model 1, Model 2 and Model 3. In this study, we have investigated these models in terms of their accuracy in diagnosing diseases that affect tea leaves. Various images of healthy and diseased tea leaves in the dataset were used to train and test the models. Model 3 achieved a 99.65% accuracy making it the most successful whereas Model 2 and the Model 1 models achieved 98.7% and 98.58% accuracy respectively using 5-fold cross-validation. Our research indicates that these models can detect tea diseases accurately and efficiently, potentially enhancing early disease identification and crop quality. Professionals can assist quickly and easily in detecting diseases in plant leaves early.

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1. Introduction

Tea is classified as a plantation crop since it is grown extensively. Tea is widely consumed and has a very long history. Researchers have also shown that tea helps prevent Alzheimer's disease and lessen anxiety. The increasing demand for tea makes protecting tea trees and leaves critical. The demand for tea is rising steadily in the modern era, but the supply is falling for several reasons, including weather and climate change. Tea is grown throughout huge

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agricultural landscapes and is a staple of many regions' economies and cultures. It is one of the most special drinks in the world. Numerous plant diseases, however, pose a danger to the sustainability of tea production since they can result in significant crop loss and serious damage to the vegetation. It is essential to identify these conditions early to put disease control plans into action, and it also protects the jobs that this industry provides. Due to the advent of precision agriculture and computational approaches, which have created new avenues for disease identification and treatment, tea production appears resilient despite these phytosanitary challenges. This research includes in-depth studies of the pathophysiology of tea leaf diseases, their importance on crop quality and productivity, and recent methods for diagnosis and treatment. Numerous diseases may reduce tea yield and result in financial loss for farmers [17]. The UN Food and Agriculture Organization (FAO) predicts a global population of 9.1 billion by 2050. Food consumption will rise as the population grows [3]. Meanwhile, it is difficult for nutrient levels to rise due to a lack of cultivable and access to clean water. The fast development of precision and agriculture in recent years has shown to the overall usage of computer image processing technology to unravel various challenges within agricultural sciences [20]. Plant species classification, disease identification, and nutrient content estimation are a few applications for these technologies [1]. The mobile application also helps the farmers to identify leaf disease and make earlier decisions [2]. Manual inspection of leaf disease is a hectic process and requires domain knowledge [18]. When farmers handle it manually, they waste a significant amount of their energy, time, and effort, which significantly reduces crop yield [5]. We proposed an effective DL technique named VGG19+CNN, (VGG19+ResNet50+CNN) using transfer learning and GAP-VGG19 presented to address the current challenge of tea leaf disease classification. By adding the hybrid approach and dense layers, as well as custom layers at the end of the framework structure, we have altered the VGG19 and ResNet50 models that were previously in use. Using an end-to-end training mechanism, the VGG19, CNN, and ResNet50 that are being presented extract high-level indicators of infected regions and associate them with corresponding classes robustly. This is accomplished through the use of transfer learning and global average pooling. Furthermore, the hybrid model approach raises the proposed method's recall power by including relevant facts about observable characteristics like diseased plant leaf areas. The CNN(Convolutional Neural Network), VGG19(Visual Geometry Group), and ResNet50(Residual Network) models were also tested and trained on the same tea leaves dataset as our proposed models to evaluate their accuracy. Additionally, The following is a summary of this research:

1. A VGG19 model with custom layers of CNN model named VGG19+CNN will be referred to as Model 1.
2. Using Transfer learning a VGG19 model with a merge with ResNet50 with a custom layer of CNN model named (VGG19+ResNet50+CNN) using transfer learning will be referred to as Model 2.
3. A VGG19 model merged with the Global average pooling named GAP-VGG19 will be referred to as Model 3.

1.1. Distinctive Contributions to Work

The research also helps the researchers to understand hybrid models and encourages them to have a deeper understanding. An effective Hybrid DL approach called Model 1, Model 2, and Model 3 is suggested that is proficient in detecting relevant and distinctive image features and shows improved tea leaf disease classification results. Transfer learning and global average pooling are adopted to enhance the capability of the proposed models. Hyperparameter tuning is used to cope with the problem of overfitting, which improves the accuracy of classifying tea leaf-infected regions. To demonstrate the proposed model's effectiveness, we compared the base models like CNN, VGG19, and ResNet50 to evaluate the classification results. The proposed approach addresses the specific **challenges** associated with the diagnosis of tea leaf diseases. Previous research has demonstrated the potential of CNNs and their variants in achieving high accuracy for general plant disease identification. This study leverages and extends these existing models by integrating advanced techniques such as transfer learning and global average pooling to improve diagnostic accuracy further. Using hybrid architectures, we combine the strengths of these pre-trained models to enhance their feature extraction capabilities and performance. This study bridges the gap in the existing literature by employing transfer learning and global average pooling, providing a tailored solution for the tea leaf disease domain. One potential **limitation** of our study is the reliance on a dataset that may not fully represent the diversity of tea leaf diseases across different geographical regions and climatic conditions. Even though the hybrid models demonstrated high accuracy on the provided dataset, it remains unclear whether they are generalizable to other crops or different types of leaf diseases. The statistical analysis, while robust within the context of the given dataset, may have limitations

in addressing class imbalances or subtle variations within disease categories. Model performance can be skewed by such imbalances, favoring more prevalent classes at the expense of rarer disease types. This targeted approach not only reduces costs associated with excessive usage but also promotes sustainable farming practices by minimizing environmental impact. The subsequent sections of this research paper are **structured** as follows: Section 2 displays the research that various researchers have done. Section 3 Image Pre-Processing and Data Augmentation. Section 4 explains the proposed model 1, model 2, model 3. Section 5 discusses the Experimental Results and section 6 discusses the statistics of Accuracy, Loss, F1-Score, AUC, and PR for Proposed Models.

2. Literature Review

Since ancient times, tea has been a popular beverage. However, it can be affected by several diseases, which could lower overall production and cause farmers to lose money on their crops. An image processing technique has been proposed by the author [7] to identify healthy leaves and algal diseases. The author employs an SVM classifier to identify the sickness in tea leaves. A deep CNN model named LeafNet was developed by author [4] which automatically extracts features using different filters of various sizes. SVM and MLP models based on dense, scale-invariant properties are utilized in a bag-of-words approach to categorize tea leaf diseases. Image processing is used in [11] to resize, crop, and convert images. The author employed a Neural Network Ensemble to recognize patterns. Lastly, the recovered features and the type of tea leaf disease are sent into the artificial neural network to train it. To identify the tea leaf disease, the authors [16] fine-tune pre-trained models like VGGNET, Xception, and ResNet on imageNet data. Additionally, the author redesigned pre-trained models using a vanilla pre-trained model. The author in [6] proposed a LeNet model based on Deep CNN (DCNN) to identify the various leaf diseases in tea leaves. There are two average pooling layers and two convolutional layers in this model. The author [13] suggested faster R-CNN to find lesions on tea leaves and identify their reasons. The suggested method could be utilized for disease identification in the field and attained an accuracy of 89.4%. The authors [24] utilized a decision tree for feature selection after acquiring the hyperspectral images. This allowed them to destroy the irrelevant 16-dimensional features. The decision tree has 80% accuracy compared to the recognizer utilized in tea leaf sickness. The authors [23] have proposed the most recent models, GAN and DCGAN, which are utilized for data augmentation in identifying tea leaf disease. These models are different from the work proposed by other researchers. The author evaluated the performance of suggested models using DenseNet, ResNet, AlexNet, and Xception DCNNs. The author [21] has worked on a well-known method known as transfer learning to identify tea diseases such as anthracnose, white tea scab, and tea blister blight. In addition, the author employed the Inception v3 convolutional layer model on a limited data set, demonstrating improved accuracy and recall compared to conventional techniques. In this study, the author [8] has suggested the Retinex algorithm as a means of improving image quality. To detect the severity of the condition, the images are sent into the VGG16 network after being enhanced using Faster RCNN. Long-term agriculture also depends on health monitoring, the author [12] suggested a CNN model with two fully connected, four convolutions, and one input layer to identify eight different forms of tea leaf diseases. The author [10] proposes a computer vision approach based on binary image processing and the HSI model to analyze tea leaf images for color and form features to grade the quality of the tea. Using a genetic neural network, tea may be identified automatically with a high degree of accuracy compared to manual tests. Though more work needs to be done on feature selection and quality quantification, the system has the potential to replace human sensory evaluation. To examine the molecular defense mechanisms of tea against blister blight disease, author [9] analyzed the transcriptome of susceptible and resistant tea genotypes using RNA-seq. They found a total of 37,790 unique transcripts, with 149 defense-related genes identified as important players. Our proposed hybrid deep learning models perform **better** than existing approaches by utilizing multiple novel techniques. First, our models take advantage of improved feature extraction capabilities by combining transfer learning with global average pooling. This improves accuracy and makes it possible for them to generalize better. The Proposed models outperform cutting-edge models when compared, with greater accuracy rates-99.65%, 98.7%, and 98.58%. The proposed approach is effective in accurately diagnosing tea leaf disease. These hybrid models show outstanding increases in efficiency, using less computational power and time for training and predictions, in addition to being technically robust, having been evaluated under numerous folds and situations. The proposed approach is especially useful in real-world applications where resource and speed limitations are crucial. Furthermore, the proposed models' capacity to adjust to new diseases without requiring significant retraining makes them a practical answer for continuous agricultural

problems, improving early disease detection capacities and raising crop output and quality. The proposed approach offers practical advantages for disease management in agriculture.

3. Image Pre-Processing and Data augmentation

The dataset has disease categories Healthy, Gray light, Brown blight named as Brown blight, anthracnose, algal leaf, red leaf spot, white spot, healthy, gray light, and bird eye spot. The total number of leaves is 885 and Tea leaves after Data Augmentation are 5310. Among the leaf images in the dataset, there are 3 classes: healthy, Gray light, and Brown blight. To train the model, the leaf images of tea plants have been randomly divided so that 80% form the training dataset, while 20% form the testing dataset. The purpose of pre-processing is to enhance the image data by lowering undesired distortions. various data augmentation technique is followed such as rotation, Width shift, Height shift, Shear, zoom, and horizontal flip. Fig. 1 shows images sample after augmentation. The actual images were 885 in the dataset and after data augmentation, the number of images became 5310.

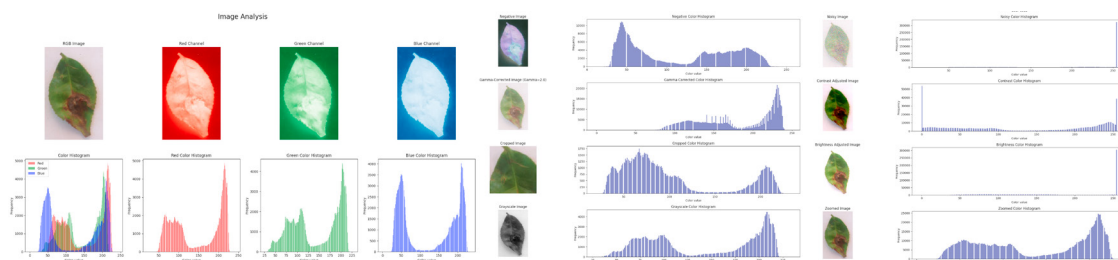


Fig. 1: Data Augmentation Techniques of RGB, Red channel, Green Channel, Blue Channel, Negative image, Gamma corrected

4. Proposed Models

There are three major models, namely Model 1, Model 2, and Model 3, that have been discussed in this section. Fig. 2 showcases the proposed methodology for the proposed models.

4.1. Model 1

Model 1 is a hybrid model based on the VGG19 architecture, to which CNN custom layers are added for distinct image categorization. With its small (3x3) convolutional filters, as demonstrated in Fig. 3, the 19-layer structure of VGG19 is well-known for its ability to extract fine-grained properties from images. This technique uses five convolutional layer blocks that use max-pooling to reduce dimensionality and fully linked layers for classification. A large ImageNet dataset was used to train the hybrid model, which is based on the pre-trained VGG19 architecture (refer Table 1). Furthermore, the model could extract features effectively through transfer learning. A task-specific dataset determines which layers to add to the model to make it adaptable to different tasks. Regularization, dropout, and batch normalization are frequently used to improve feature processing. After the first processing step, the model extracts and compresses features from the input image using convolutional and pooling layers. Once the data has passed through these layers, it is compressed and introduced into dense layers for categorization. Dropout layers reduce overfitting by deleting some connections at random during training. Algorithm 1 describes the complete process of Model 1.

4.2. Model 2

The layered 3x3 convolutional filters and depth of the deep CNN, VGG19 are well known. This model exceptionally well captures fine-grained patterns in image data. On the other hand, ResNet50 solved the vanishing gradient issue by introducing the idea of residual learning with skip connections, which made it possible to train much deeper networks. The number "50" in ResNet50 indicates that the design has 50 layers. This model 2 as shown in Fig. 4 leverages the unique qualities of two distinct architectures (refer Table 1). Together, ResNet50's abstract feature extraction and VGG19's capacity for fine detail recognition provide a strong basis for precise image classification. By leveraging pre-existing knowledge, a model can significantly accelerate training and enhance performance on a target

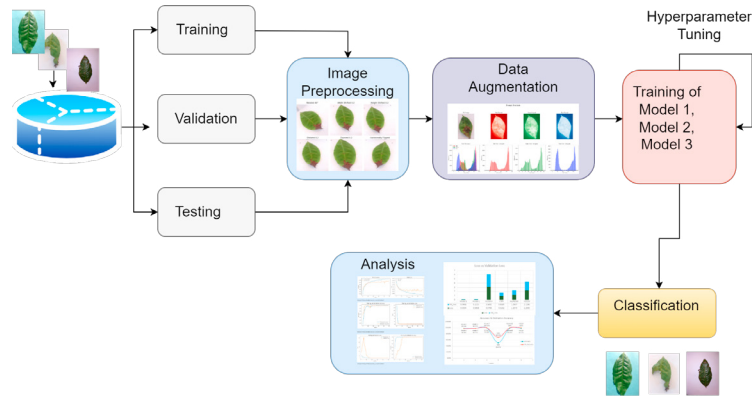


Fig. 2: Proposed methodology

Algorithm 1 Tea leaf disease detection using Model 1

Require: Training dataset $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$

Ensure: Trained parameters θ

- 1: Load pre-trained VGG19 (without top), set layers to non-trainable.
 - 2: Append custom dense layers on top for classification: Flattened layer, Dense layers, Dropout layer
 - 3: Compile model with Adam, categorical cross-entropy, accuracy.
 - 4: Define ImageDataGenerators for data preprocessing and augmentation: Rescale pixel values to $[0, 1]$, apply rescaling, random rotations, shifts, shear, zoom, flip
 - 5: Create data iterators using *flow_from_directory*.
 - 6: Implement callbacks for EarlyStopping and ReduceLROnPlateau.
 - 7: Train model using *fit*, validate with validation data.
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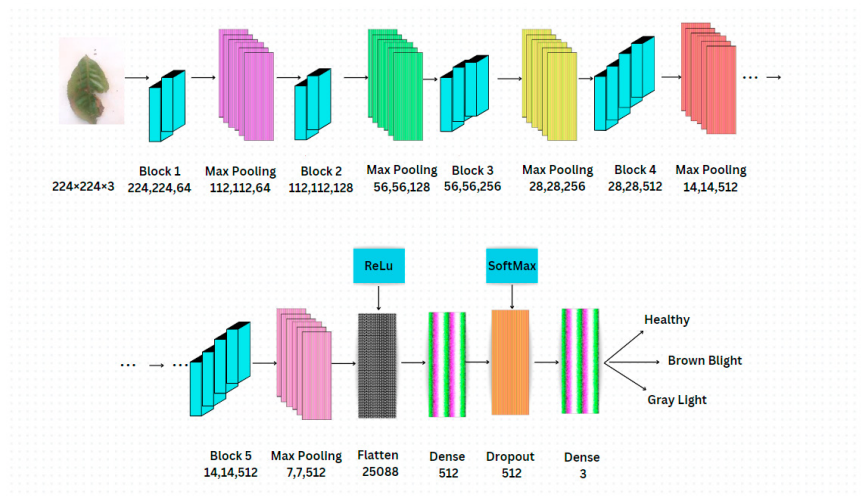


Fig. 3: Model 1 Architecture

dataset, even with limited training data. This is achieved through transfer learning. Algorithm 2 describes the complete process of Model 2.

4.3. Model 3

The proposed model is a high-capacity neural network that handles challenging image categorization applications. It uses pooling layers to lower dimensionality and impose spatial hierarchies, and deep stacks of convolutional layers to learn a hierarchy of features. In comparison to fully connected layers, the global average pooling layer dramatically

Algorithm 2 Tea leaf disease detection using Model 2

Require: Training dataset $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$

Ensure: Trained parameters for the hybrid CNN model

- 1: Initialize input layer with shape (224, 224, 3)
 - 2: Load pre-trained VGG19 and ResNet50, exclude top layers:
 - a: Apply VGG19 and ResNet50 to input layer
 - b: Flatten output
 - 3: Concatenate flattened outputs from VGG19 and ResNet50
 - 4: Append dense and dropout layers:
 - Dense layer with 1024 neurons and ReLU activation and Dropout layer with 0.5 dropout rate
 - 5: Compile model with Adam optimizer (lr = 0.0001) and categorical cross-entropy loss
 - 6: Use the trained model and evaluation results for classification
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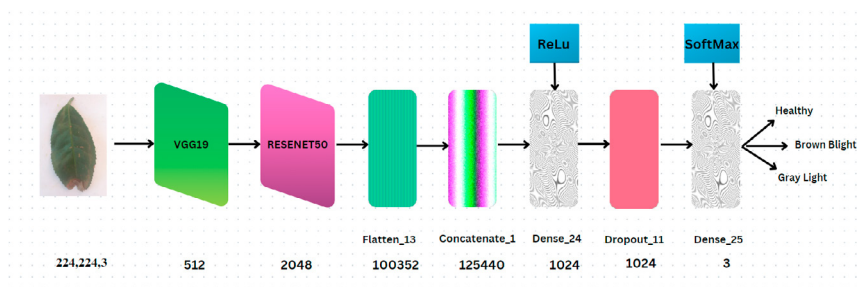


Fig. 4: Model 2 Architecture

minimizes the number of parameters and serves as a structural regularize. When dealing with a large number of parameters, the suggested design, as illustrated in Fig. 5, helps decrease overfitting and boosts the model's efficiency. The dropout layer enhances the model's ability to generalize to new, unseen data by preventing an over-reliance on a single activation route. The final dense layer applies the learned characteristics to the classification targets, allowing the model to identify the leaf sickness (refer Table 1). Algorithm 3 describes the complete process of Model 3.

Algorithm 3 Tea Leaf Disease Detection Model 3

Require: Training dataset $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$

Ensure: Trained VGG19 model for classification

- 1: Load VGG19 model pre-trained on ImageNet without the top layer and Add Global Average Pooling (GAP) layer
 - 2: Add a Dense layer for classification with softmax activation and 3 classes and Compile the model.
 - Rescale by 1/255
 - Random transformations: rotation, width shift, height shift
 - Shear, zoom, horizontal flip, fill mode 'nearest'
 - 3: Initialize ImageDataGenerator for validation data with rescaling by 1/255
 - 4: Load training data from directory:
 - Target size (224, 224), batch size 32, class mode 'categorical'
 - 5: Load validation data similar to training data setup
 - 6: Train model using `model.fit` with specified steps per epoch and validation steps
 - 7: Use the trained model for classification
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5. Experimental Results

In this section, we will discuss the outcomes of the proposed models, which were trained and evaluated using a tea leaf dataset. To assess the performance of the proposed hybrid Model 1, Model 2, and Model 3, a 5-fold cross-validation was performed and the model training was set for 50 epoch cycles, a batch size of 8, reduce_lr with 0.2 factor where the min_lr is 1e-5 with the early stopping criteria of 30 patience. Fig. 6 displays the accuracies and loss on the training and validation dataset for the proposed models.

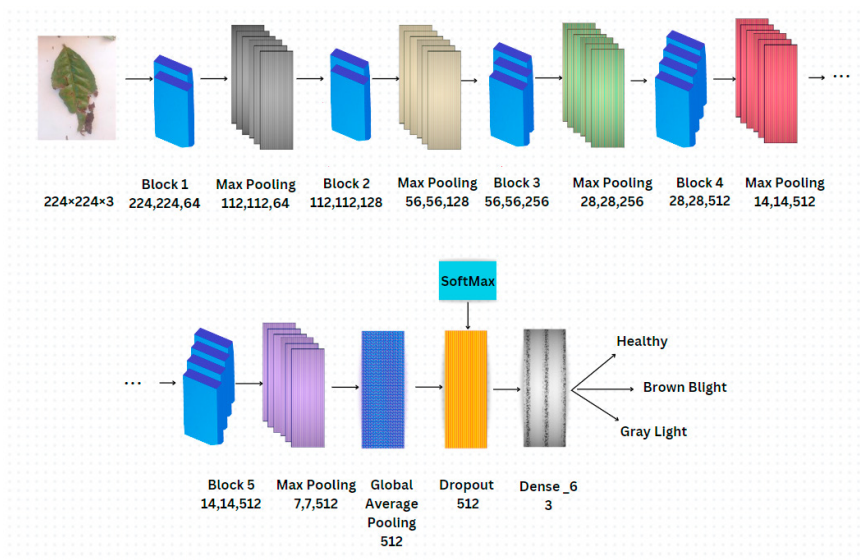


Fig. 5: Model 3 Architecture

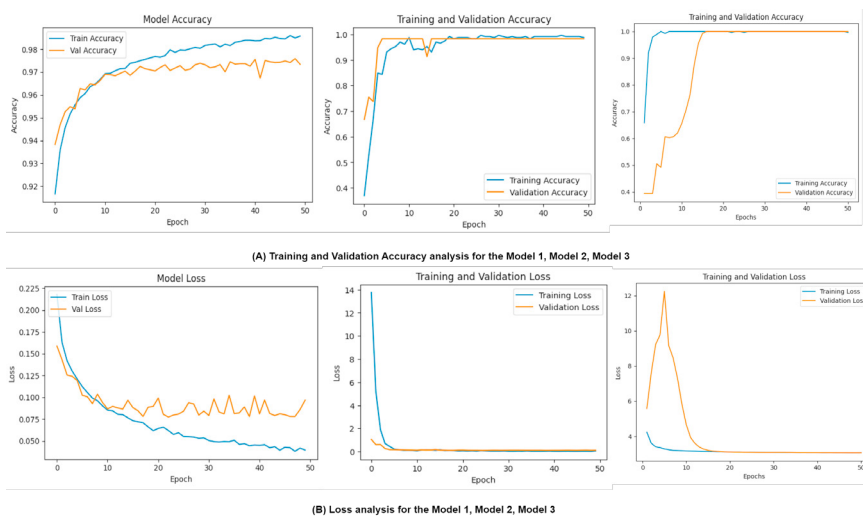


Fig. 6: Graph of Training and Validation Accuracy vs. loss for (A) Model 1 (B) Model 2 (C) Model 3

6. Discussion

This section will discuss our proposed and existing model based on accuracy, loss, AUC-ROC, confusion matrix, F1 score, and precision recall. Feature extraction from images is an important step in training deep-learning models because the quality of the extracted features determines the model's performance.

6.1. Accuracy Comparison of Models

Initially, some existing DL models such as CNN, VGG19, and, ResNet50 are implemented on the tea leaf dataset to make a comparative analysis with the proposed models. Further, these existing models were compared with our proposed hybrid models on the tea leaf dataset using the implementation details as mentioned in Table 2. The proposed Hybrid model outperformed the other model in terms of accuracy, as shown in Fig. 6 and 7 (A). In comparison to the accuracy of CNN (68.42.00%), VGG19 (98.27%), and ResNet50 (98.48%), Model 3 achieved the maximum accuracy of 99.65%, Model 1 achieved an accuracy of 98.58%, and Model 2 reached 98.70% accuracy. Model 3 outperformed

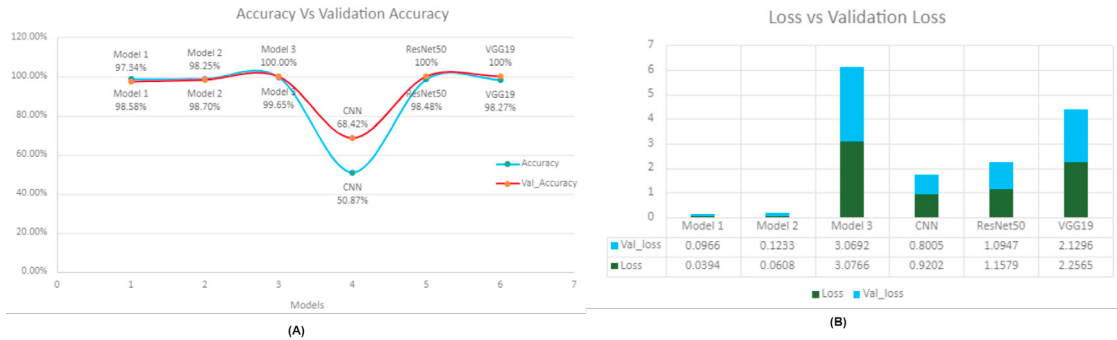


Fig. 7: (A) Accuracy and (B) Loss analysis of the proposed model in comparison to base models.

both Model 1 and Model 2 with 1.07% and 0.95% better accuracy, respectively, making it the best proposed model. The validation accuracy of the proposed models (97.34%, 98.25%, 100% respectively) shows a significantly better performance than the base models (50.87%, 100%, 100%). The proposed model was also compared with the current studies that use ML and DL models for tea disease identification. The most satisfactory machine-based methods were YOLOv7 and Resnet50. In the DL-based method, the proposed models exceeded all of the current ones. Fig. 7 the comparative analysis of the proposed method with the existing studies.

Table 2: Comparative analysis of the Proposed method with the existing research

Author	Model	Accuracy
[22]	ResNet-50 and Inception V3	Accuracy 97.44%, 97.62%.
[17]	YOLOv7	Accuracy 97.3%
[19]	CNN	Accuracy 84%
[15]	DCNN	Accuracy 98.99%
[14]	SVM	Accuracy 83%
[6]	LeNet	Accuracy 90.23%
[16]	VGGNet, ResNet, and Xception	Accuracy 84.86%, 94.05%
[4]	LeafNet algorithm, SVM, MLP algorithm	Accuracy 90.1%, 60.62%, 70.77%
Trained on Tea leaf Dataset	CNN	Accuracy 50.87%
Trained on Tea leaf Dataset	ResNet50	Accuracy 98.48%
Trained on Tea leaf Dataset	VGG19	Accuracy 98.27%
Proposed Model	Model 1, Model 2, Model 3	Accuracy 98.58%, 98.7%, 99.65%

6.2. Loss Comparison of Models

Initially, we have implemented some existing DL techniques such as CNN, VGG19, and, ResNet50 on the tea leaf dataset. Further, these existing models were compared with our proposed hybrid models as shown in Fig. 7 (B). The model 3 is the most successful of all the models presented. The model's ability to correctly predict the data increases with decreasing loss. VGG19 model performs the lowest when compared to the other models with loss (2.2565%). A larger loss suggests that the model forecasts are not as precise. Model 2 with a loss of just 0.0608 % is operating well through transfer learning and appears to have greatly increased its performance. The model 1 loss of 0.0394% also indicates strong performance, however not as excellent as the models with transfer learning or global average pooling. A loss of 0.9202% is recorded by the CNN model, which is lower than that of the VGG19 and ResNet50 models but noticeably higher than that of the hybrid models including VGG19. ResNet50 with a loss of 1.1579% percent performs in the lower-middle range of the models mentioned.

6.3. F1-Score of Proposed Models

Model 3 and model 2 precision and recall are both flawless for each class, demonstrating the perfect balance between classes and properly identifying all important instances of each class. Each bar appears to be at its maximum

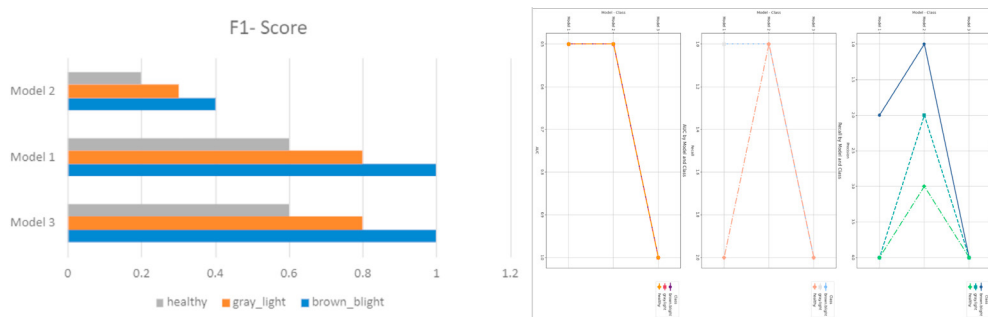


Fig. 8: F1 Score, AUC and PR for proposed models

height, indicating that the model has achieved flawless classification performance for every class on the dataset. None of the classes get the maximum score of 1 for Model 2 (Fig. 8). The F1 scores of the brown_blight and gray_light classes are both above 0.3 in comparison to the healthy class.

6.4. AUC and PR for Proposed Models

Model 3 performs exceptionally well in classifying each class with an AUC of 1.00. This model's great accuracy is due to the use of Global Average Pooling. Meanwhile, the PR curve for Model 3 shows normal model performance and has a lot more variability. The 'gray_light' and 'brown_blight' classes show a significant decrease in precision as recall rises. Model 2 AUC is 1.00 for all three classes showing that the model differentiates between the positive and negative classes for each category. The PR curve for Model 2 for all three classes starts high (almost 1.00) but decreases as recall rises. The Model 1 AUC varies greatly between the classes. The 'healthy' and 'gray_light' classes have AUCs (Fig. 8) of about 0.50. The AUC for the 'brown_blight' class is 0.51, which shows the model is not significantly performing. The PR of Model 1 for the three classes—brown_blight, gray_light, and healthy—signifies perfect recall and precision scores of 1.00. Model 3 and Model 2 demonstrate perfect class classification.

7. Conclusion and Future Scope

This research aimed to analyze and apply proposed hybrid deep-learning models to diagnose diseases that damage tea leaves. The specialized DL architectures VGG19, CNN, and ResNet50 employing transfer learning, and global average pooling models were combined with three new hybrid models for disease prediction and classification in tea plant leaves. The study's findings support the notion that DL models—such as Model 1, Model 2, and Model 3 are useful for identifying tea leaf disease. Out of all these models, Model 3 has the highest accuracy on the dataset 99.65%. Model 3 outperformed Model 1 in terms of accuracy (98.58%) and Model 2 (98.7%) while still demonstrating good accuracy. According to these results, DL models may be able to detect tea leaf disease with accuracy and efficiency, ultimately aiding the agriculture industry. Based on the analysis results, it was determined that the suggested model outperformed other models. The approach for diagnosing tea-related diseases that have been put out is novel. The technology hasn't been included in a mobile application, which is one of the major limitations and further scope of this work. Nevertheless, because it only needs the user to take images of the afflicted leaf, it does offer a quick and inexpensive way to diagnose tea leaf diseases. By utilizing more sophisticated AI techniques that are enabled by Internet of Things (IoT) technology, we hope to improve the model in the future.

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